

Average Cost by Region

Based on the information available and the queries conducted, the top three regions with the highest average cost are the Galar, Johto, and Kanto regions. The Galar and Johto stand out compared to the rest of the regions as the other regions all have average visit costs within \$100 of each other but Johto has an average cost close to \$200 above the third highest cost (Kanto) while Galar has an average cost of close to \$300 higher than Kanto. While Galar stands far above the other regions in cost this does not reflect real life metrics as all the data generated (aside from the name of the patient) is synthetic. Since this project uses synthetically generated visit costs and diagnoses, these differences should not be interpreted as evidence that patients from these regions require more expensive healthcare in real life. Instead, the results demonstrate how SQL can be used to summarize and compare healthcare cost data across different patient groups. If this were a real healthcare dataset, additional analysis would be required to determine whether factors such as diagnosis, visit type, age, insurance status, or other patient characteristics contributed to the higher average costs.

Average Cost by Visit Type

Since emergency visits are both the most frequent and most expensive visit type, how do average costs differ by visit types? By using the information gathered it is very clear that the most expensive visit type by far are emergency treatments. The average cost of an emergency visit is \$1720.24 which is almost \$1300 more than the second most expensive visit, specialist at \$476.01. It is important to note that while this emergency has the highest average cost, it does not link it to any one specific region as such no conclusion can be made about which region has the most emergency visits or which regions are the most expensive with regards to visit types. A question that can be brought up by analyzing this query is what percentage do emergency visits contribute to this facility's total visit costs?

Average Treatment Delay

This query shows the average number of days between diagnosis and treatment for each condition, as well as the number of patients with that condition. One limitation is that the average by itself does not show the full distribution of treatment delays. Adding the minimum and maximum delay could help identify whether certain patients waited significantly longer than others. With real healthcare data, additional information such as scheduling, staffing, treatment requirements, and day of the week could also help determine why some conditions have longer treatment delays.

Diagnosis Count by Region

What does the diagnosis condition type disbursement look like per region? This query shows each region and what conditions patients have who live there and how many. Because diagnoses were randomly assigned during data generation, differences in diagnosis counts between regions should not be interpreted as evidence of regional differences in disease prevalence. With real healthcare data, however, this type of analysis could be used to identify geographic differences in disease burden and determine which populations may require additional investigation or resources.

Followup Visits

This query was created to identify records labeled as follow-up visits and verify that the follow-up occurred after the patient's diagnosis date. This acts as a data-quality check because a follow-up visit occurring before the diagnosis would violate the rules used when generating the synthetic dataset. In a real healthcare database, similar validation queries could be used to identify records with inconsistent dates that may require further review.

Patient's Care Record

The main purpose of this query is just to see how the JOIN functions interact and work by combining all the information of the patients into one cohesive table for easier viewing and analysis.

Top 5 Expensive Visits

The purpose of this was to see which visits were the most expensive of all the patients. All of them were emergencies with costs of at least \$2300. One worthwhile note is that two entries are from Galar which bring up another question, does Galar have a much larger proportion of emergency visits compared to other regions? Further investigation is required to answer this question.